Pons Nelly

PhD Candidate • Mechanical & Materials Engineering • Defence: April 2026

Toulon, France • nelpons@gmail.com • +33 (0)6 51 69 48 84 • nellypons.wordpress.com
Driving licence B • FR French nationality • Age: 30

WHY ME?

PhD candidate in Materials and Mechanical Engineering specialising in additive manufacturing and multi-scale characterisation of piezoelectric PVDF. Strong experience in experimental testing and numerical modelling (FEM). International background across Europe and North America. Seeking an engineering role in a high-performance R&D environment.

SKILLS

Materials, Mechanics & Simulation

Solid/fluid mechanics, FEM, structural dynamics, vibroacoustics, materials science (polymers, composites)

XRD, FTIR, DSC, SEM, tomography, synchrotron (SOLEIL)

Software, Programming & Data

Abaqus, COMSOL, Ansys, CATIA V5, SolidWorks, AutoCAD, HyperMesh, VA One

Python, C++, Matlab, VBA, SQL, LabVIEW, LaTeX, MS Office & Project

Management & Softskills

Project Management (Gantt, Pert), technical communication, teamwork, Problem solving, Adaptability / fast learning

LANGUAGES

French native

English fluent

German, Dutch, Spanish, Italian basics

INTERESTS

Aeronautical licensing intro (BIA), table tennis (player, referee, umpire), hiking, photography, travel, crochet

EXPERIENCE

2021 - 2026 (4.5 yrs) **PhD Student • Ecole des Mines Paris PSL • Paris, FR**
Additive manufacturing of PVDF composites: investigation of process-structure-property relationships • Mechanical testing to analyse deformation mechanisms and phase transitions • In-situ synchrotron experiments at SOLEIL Synchrotron (XRD, tomography under mechanical loading) • Experimental data analysis to identify mechanisms governing mechanical behaviour • Teaching assistant, Materials Science (graduate level)

2024 **Volunteer • Olympics & Paralympics Games • Paris 2024**

2020 - 2021 (1 yr) **Young Graduate Trainee • European Space Agency (ESA) • Noordwijk, NL**
Structural analysis, vibroacoustic in-/out-of-plane responses (VA One), test campaign (sinusoidal & random vibrations)

2019 - 2020 (6 mths) **Intern • Airbus Defence and Space • Friedrichshafen, DE**
Review and work on laser surface treatment of various materials for Space applications

2018 - 2019 (10 mths) **Research assistant• CurAL Lab, UQAC • CA**
Preparation and analysis of hydrophobic surfaces

2018 (3 mths) **Intern • Materials Sc. Lab, Univ. Politehnica • Bucharest, RO**
Creation and extrusion of a new filament with CaCO3 and ABS

2017 (1 mth) **Intern • Geosciences Laboratory, Univ. Lyon 1 • FR**
Experiment design and execution using LabView, Matlab programming with MatPiv for velocity field analysis

EDUCATION

2026 **PhD Mechanical and Materials Science • Mines Paris PSL • FR**

2020 **M. Sc. of Engineering spe. Materials Science • UQAC • CA**
2020 **M. Sc. of Physics Engineering • Polytech Clermont-Ferrand • FR**

2016 Preparatory level for Eng. School • Polytech Lyon • FR
First year, medicine school • Univ. Lyon 1 • FR

2013 Baccalauréat • Military High school • Aix-En-Provence, FR

AWARDS

2018 **Winning team • Innovation for the Nuclear I4N Competition, www.iync.org/i4n/ • Bariloche, AR**

REFERENCES

PhD Directors - Mines Paris PSL:

Prof. K. Inal (karim.inal@minesparis.psl.eu) • Prof. L. Laiarinandrasana (lucien.laiarinandrasana@minesparis.psl.eu)

Letters of recommendation available:

M. Sc. G. Coe (YGT Supervisor ESTEC, ESA, Noordwijk, NL); **M. Sc. M. Süss** (Internship Supervisor Airbus Defence and Space, Friedrichshafen, DE); **Assoc. Prof. H. De Baynast** (Prof. at Polytech Clermont-Ferrand, Univ. Auvergne, FR); **PhD T. Alboussière** (Geosciences Laboratory, Univ. Lyon 1, FR); **Assoc. Prof. D. Batalu** (Materials Science Laboratory, Univ. Politehnica, RO)

20.03.2026

Publications & Conference Presentations

PHD THESIS

Pons Nelly (2026). "From 3D-Induced Crystallisation and Structural Organisation to the Control of Electromechanical Anisotropy and Damage in Piezoelectric PVDF." Université Paris Sciences et Lettres PSL.

Defence on 28th April 2026

PUBLICATIONS

Pons Nelly, Bafétégué L. Ouattara, Séverine A.E. Boyer, Alain Burr, Lucien Laiarinandrasana, Cristian Ovalle, Karim Inal (2026). "Structure and Crystallization Kinetics of PVDF/Organoclay Nanocomposites Processed by Twin-Screw Extrusion and Electrical Contact Poling." *Materials Today Communications,* 50, 114535.

DOI: https://doi.org/10.1016/j.mtcomm.2025.114535

CONFERENCES

Oral presentations

Pons Nelly, Séverine A.E. Boyer, Alain Burr, Cristian Ovalle, Karim Inal, Lucien Laiarinandrasana (2024). "Effects of clay filler contents and a high voltage electric field on the microstructure and mechanical response of 3D-printed PVDF composites."

SEMTEAM Seminar, Mines Paris PSL, Evry • 17th Int. Conf. on Advanced Computational Engineering and Experimenting (ACEX), Barcelona, Spain

Pons Nelly, Karim Inal, Séverine A.E. Boyer, Alain Burr, Lucien Laiarinandrasana, Cristian Ovalle (2023). "3D FDM printing and electric poling for a smart anisotropic piezoelectric PVDF."

17th European Conference on Spacecraft Structures Materials and Environmental Testing (ECSSMET), Toulouse, France

Pons Nelly, Séverine A.E. Boyer, Alain Burr, Karim Inal (2022). "Multiscale Considerations of the Anisotropic Piezoelectric Behaviour of the β Phase PVDF."

World Conference on Materials Science and Nanotechnology (WCMSN), Bangkok, Thailand *(online oral presentation)*

Poster presentations

Pons Nelly, Karim Inal, Séverine A.E. Boyer, Alain Burr, Lucien Laiarinandrasana, Cristian Ovalle (2023). "3D FDM printing and electric poling for a smart anisotropic piezoelectric PVDF."

17th European Conference on Spacecraft Structures Materials & Environmental Testing (ECSSMET), Toulouse, France

Pons Nelly, Séverine A.E. Boyer, Alain Burr, Karim Inal (2022). "Multiscale Considerations of the Anisotropic Piezoelectric Behaviour of the β Phase PVDF."

CRISTECH, Ecully, France